

COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	9/30/2026	DATE SUBMITTED:	10/05/2026
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ACTIVITY NO. 10.1
Install, Configure, and Manage Log Monitoring tools

1. PROCEDURE OUTPUT

1. Create a playbook that:
- a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)

Base roles

GNU nano 6.2
main.yml

```

--
- name: Debian family setup
  when: ansible_os_family == "Debian"
  block:
    - name: Install prerequisites
      ansible.builtin.apt:
        name: [apt-transport-https, gnupg, curl]
        state: present
        update_cache: true

    - name: Ensure keyrings directory exists
      ansible.builtin.file:
        path: /etc/apt/keyrings
        state: directory
        mode: '0755'

    - name: Download Elastic GPG key
      ansible.builtin.get_url:
        url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
        dest: /etc/apt/keyrings/elastic.asc
        mode: '0644'

    - name: Add Elastic APT repository
      ansible.builtin.apt_repository:
        repo: "deb [signed-by=/etc/apt/keyrings/elastic.asc] https://artifacts.elastic.co/packages"
        state: present

```

^G Help
^O Write Out
^W Where Is
^K Cut
^T Execute
^C Location
^X Exit
^R Read File
^_ Replace
^U Paste
^J Justify
^_ Go To Line

Elasticsearch tasks roles

```
GNU nano 6.2 main.yml *
--
- name: Install Elasticsearch
  ansible.builtin.package:
    name: elasticsearch
    state: present

- name: Configure Elasticsearch
  ansible.builtin.template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
  notify: Restart elasticsearch

- name: Enable and start Elasticsearch
  ansible.builtin.service:
    name: elasticsearch
    state: started
    enabled: true

- name: Open port 9200 (CentOS firewalld)
  ansible.posix.firewalld:
    port: 9200/tcp
    permanent: true
    immediate: true

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Elasticsearch handler roles

```
GNU nano 6.2 main.yml *
--
- name: Restart elasticsearch
  ansible.builtin.service:
    name: elasticsearch
    state: restarted
```

Elasticsearch template roles

```
GNU nano 6.2 roles/elasticsearch/templates/elasticsearch.yml.j2
cluster.name: elastic-lab
node.name: {{ inventory_hostname }}
path.data: /var/lib/elasticsearch
path.logs: /var/log/elasticsearch
network.host: 0.0.0.0
discovery.type: single-node
xpack.security.enabled: false
xpack.security.transport.ssl.enabled: false
xpack.security.http.ssl.enabled: false
```

Kibana tasks roles

```
GNU nano 6.2                                main.yml *
```

```

--
- name: Install Kibana
  ansible.builtin.package:
    name: kibana
    state: present

- name: Configure Kibana
  ansible.builtin.template:
    src: kibana.yml.j2
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: '0660'
  notify: Restart kibana

- name: Enable and start Kibana
  ansible.builtin.service:
    name: kibana
    state: started
    enabled: true

- name: Open port 5601 (CentOS firewalld)
  ansible.posix.firewalld:
    port: 5601/tcp
    permanent: true
    immediate: true

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line

```

Kibana handler roles

```
GNU nano 6.2                                main.yml *
```

```

--
- name: Restart kibana
  ansible.builtin.service:
    name: kibana
    state: restarted

```

Kibana template roles

```
GNU nano 6.2                                kibana.yml.j2 *
```

```

server.port: 5601
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://{{ es_host }}:9200"]

```

Logstash tasks roles

```
GNU nano 6.2 main.yml *
--
- name: Install Logstash
  ansible.builtin.package:
    name: logstash
    state: present

- name: Deploy Logstash pipeline
  ansible.builtin.template:
    src: logstash.conf.j2
    dest: /etc/logstash/conf.d/logstash.conf
    owner: root
    group: logstash
    mode: '0644'
  notify: Restart logstash

- name: Enable and start Logstash
  ansible.builtin.service:
    name: logstash
    state: started
    enabled: true

- name: Open port 5044 (CentOS firewalld)
  ansible.posix.firewalld:
    port: 5044/tcp
    permanent: true
    immediate: true

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Logstash handlers roles

```
GNU nano 6.2 main.yml
--
- name: Restart logstash
  ansible.builtin.service:
    name: logstash
    state: restarted
```

Logstash template roles

```
GNU nano 6.2 logstash.conf.j2
input {
  beats {
    port => 5044
  }
}

output {
  elasticsearch {
    hosts => ["http://{{ es_host }}:9200"]
    index => "logstash-%{+YYYY.MM.dd}"
  }
}
```

2. Apply the concept of creating roles.

3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA10.1$ ansible-galaxy collection install ansible.posix:1.5.4 --force
Starting galaxy collection install process
Process install dependency map
Starting collection install process
Installing 'ansible.posix:1.5.4' to '/home/keliste/.ansible/collections/ansible_collections/ansible/posix'
Downloading https://galaxy.ansible.com/api/v3/plugin/ansible/content/published/collections/artifacts/ansible-posix-1.5.4.tar.gz to /home/keliste/.ansible/tmp/ansible-local-17217c7in35qz/tmp_stv8cbc
ansible.posix (1.5.4) was installed successfully
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA10.1$
```

```
TASK [logstash : Enable and start Logstash] *****
changed: [192.168.56.104]

TASK [logstash : Open port 5044 (CentOS firewall)] *****
skipping: [192.168.56.104]

RUNNING HANDLER [logstash : Restart logstash] *****
changed: [192.168.56.104]

PLAY RECAP *****
192.168.56.103 : ok=9 changed=0 unreachable=0 failed=0 skipped=3 rescued=0 ignored=0
192.168.56.104 : ok=10 changed=3 unreachable=0 failed=0 skipped=3 rescued=0 ignored=0
192.168.56.105 : ok=8 changed=0 unreachable=0 failed=0 skipped=4 rescued=0 ignored=0
keliste@Keliste-Workstation:~/CPE212_KELISTE/HOA10.1$
```

```
192.168.56.105 | CHANGED | rc=0 >>
PRETTY_NAME="CentOS Stream 9"
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-10-03 00:22:44 PST; 28min ago
     Docs: https://www.elastic.co
   Main PID: 7678 (java)
```

```
192.168.56.103 | CHANGED | rc=0 >>
PRETTY_NAME="Ubuntu 22.04.5 LTS"
● kibana.service - Kibana
   Loaded: loaded (/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-10-03 00:34:59 +08; 17min ago
     Docs: https://www.elastic.co
   Main PID: 6015 (MainThread)
```

```
192.168.56.104 | CHANGED | rc=0 >>
PRETTY_NAME="Ubuntu 22.04.5 LTS"
● logstash.service - logstash
   Loaded: loaded (/lib/systemd/system/logstash.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-10-03 00:46:54 +08; 6min ago
     Main PID: 5970 (java)
      Tasks: 40 (limit: 4598)
```

5. Make sure to create a new repository in GitHub for this activity

https://github.com/AiconKeliste/CPE212_KELISTE/tree/main/HOA10.1

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of having log monitoring tool?

- Having a log monitoring tool lets me see what's happening across my servers in one place instead of checking each machine's log files one by one. Also, it helps catch errors, failed services, or suspicious activity early, so problems get fixed before they turn into bigger issues. While this was only a lab setup, I can see how it's useful for security, performance tracking, and keeping a record of what happened on the system.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude Ai, Chatgpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error explanation and Command checking**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I asked claude ai for some errors that I have been encountered.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.